

## IFN647 Week 5 Review Questions

### Question 1 (Language Models)

Let  $Q = \{\text{US, ECONOM, ESPIONAG}\}$  be a query, and  $C = \{D_1, D_2, D_3, D_4, D_5, D_6\}$  be a collection of documents, where

$$\begin{aligned}D_1 &= \{\text{GERMAN, VW}\} \\D_2 &= \{\text{US, US, ECONOM, SPY}\} \\D_3 &= \{\text{US, BILL, ECONOM, ESPIONAG}\} \\D_4 &= \{\text{US, ECONOM, ESPIONAG, BILL}\} \\D_5 &= \{\text{GERMAN, MAN, VW, ESPIONAG}\} \\D_6 &= \{\text{GERMAN, GERMAN, MAN, VW, SPY}\}\end{aligned}$$

Jelinek-Mercer smoothing method uses the following equation to calculate a score for a document  $D$

$$P(Q|D) = \prod_{i=1}^n p(q_i|D)$$

where  $n$  is the number of query terms in  $Q$ , and

$$p(q_i|D) = (1 - \lambda) \frac{f_{q_i,D}}{|D|} + \lambda \frac{c_{q_i}}{|C|}$$

for all query term  $q_i$  in  $Q$ , where  $f_{q_i,D}$  is the number of times query word  $q_i$  occurs in document  $D$ ,  $|D|$  is the number of word occurrences in  $D$ ,  $c_{q_i}$  is the number of times query word  $q_i$  occurs in collection  $C$ , and  $|C|$  is the total number of word occurrences in collection  $C$ . Assume parameter  $\lambda = 0.4$ , calculate  $P(Q|D_2)$ . You are required to provide the calculation process.

### Question 2 (Likelihood Model)

Test the function `likelihood_IR(I, Q)` introduced in the lecture notes using a Jupyter Notebook. First, implement the function `likelihood_IR(I, Q)` as defined in the lecture notes. Then, apply this function to the data in the “Test\_data” folder and a query  $Q = \text{share bank}$ . The output should be a ranked list of all documents in the folder, ordered by their likelihood scores with respect to the query  $Q$ .

**Question 3** Which of the following are False? and justify your answer.

- (1) For a very large data collection, pooling technique is used to select top-k results from the rankings obtained by different retrieval algorithms. The results are merged into a pool, duplicates are removed, and the documents are presented in some random order to the people doing the relevance judgments.
- (2) Pooling produces a large number of relevance judgments for each query. However, this list is incomplete, and for a new retrieval algorithm that had not contributed documents to the original pool, this could potentially be a problem.
- (3) Many user actions (e.g., Query log data) can be considered implicit relevance judgments. The main drawback with this data is that it is not as precise as explicit relevance judgments.
- (4) A typical query log does not contain user identifier or user session identifier because of the privacy issue.

**Question 4 (Effectiveness Metrics)**

**Assumption:** Table 1 presents the relevance judgments for documents 110–124 with respect to topic 101, where a value of **1** indicates that the document is relevant and **0** indicates that it is non-relevant.

Table 2 reports the output of IR Model 1, in which documents are ranked in descending order according to their assigned weights.

**Table 1.** Relevant Judgements

| TOPIC | DocNo | Rel |
|-------|-------|-----|
| 101   | 110   | 1   |
| 101   | 111   | 1   |
| 101   | 112   | 0   |
| 101   | 113   | 1   |
| 101   | 114   | 1   |
| 101   | 115   | 0   |
| 101   | 116   | 1   |
| 101   | 117   | 1   |
| 101   | 118   | 0   |
| 101   | 119   | 1   |
| 101   | 120   | 0   |
| 101   | 121   | 0   |
| 101   | 122   | 0   |
| 101   | 123   | 1   |
| 101   | 124   | 0   |

**Table 2.** IR Model 1's Output (ranked by weight)

| TOPIC | DocNo | Weight |
|-------|-------|--------|
| 101   | 111   | 0.812  |
| 101   | 112   | 0.790  |
| 101   | 113   | 0.689  |
| 101   | 110   | 0.542  |
| 101   | 114   | 0.496  |
| 101   | 119   | 0.455  |
| 101   | 115   | 0.321  |
| 101   | 122   | 0.300  |
| 101   | 118   | 0.278  |
| 101   | 116   | 0.234  |
| 101   | 123   | 0.211  |
| 101   | 121   | 0.201  |
| 101   | 120   | 0.189  |
| 101   | 117   | 0.176  |
| 101   | 124   | 0.123  |

↑  
The set of  
retrieved  
documents  
↓

**Assumption:** Let  $A$  denote the set of relevant documents for topic 101, and  $A'$  denote the set of non-relevant documents. We further assume that IR Model 1 selects the top-6 ranked documents as the retrieved (relevant) set  $B$  (see Table 2), while  $B'$  represents the set of documents that are not retrieved.

- (1) List the elements of all sets ( $A$ ,  $A'$ ,  $B$  and  $B'$ ) and enclose them in braces.
- (2) Calculate recall and precision of IR model 1.
- (3) Calculate false positive and false negative of IR model 1.
- (4) Calculate the  $F_1$  measure of IR model 1.